1. A survey taken in Year 10 (120 students), showed the following preferences for magazines: 45 students preferred *Girlfriend*, 53 *Young at heart*, and 22 preferred *Motor sports*.

   (a) Complete the table to show this information, including the relative frequency of each magazine.

   (b) Based on this probability model, if a student was chosen at random, what is the probability that he/she would prefer
   - (i) *Motor sports*
   - (ii) **Not** *Motor sports*
   - (iii) *Girlfriend* **or** *Young at heart*

	Frequency	Relative frequency
Girlfriend		
Young at heart		
Motor sports		

(b) (i) P(*Motor sports*) = 11/60

(ii) P(**Not** *Motor sports*) = 1 - 11/60 (since '*Motor sports*' and '**not** *Motor sports*' are complementary events)
= 49/60

(iii) P(*Girlfriend* **or** *Young at heart*) = 3/8 + 53/120 since these are mutually exclusive, we can simply add their individual probabilities
= 98/120
= 49/60

(a)

	Frequency	Relative frequency
Girlfriend	45	45/120 = 3/8
Young at heart	53	53/120
Motor sports	22	22/120 = 11/60

2

Ayanda is trying on a skirt. The shop has skirts in the following sizes:

Size 7 – 20%
Size 8 – 38%
Size 9 – 30%
Size 10 - 12%

(a) Ayanda is a size 9. If there are 200 skirts in the shop, how many would fit her?

(b) If she chose any skirt at random,

(i) what is the probability that it would fit her?
(ii) what would be the **most likely** size that she would pick?

(a) Ayanda is a size 9.
30% of all the skirts in the shop are size 9. So, Ayanda would fit into 30% of 200 skirts
= 30/100 x 200
= 60 skirts

(b) (i) 30% of the skirts are Ayanda's size. Probability can be stated as a percentage. So, P(size 9) = 30% (or 3/10, or 0.3)

(ii) The highest number of skirts in the shop are size 8, so if a skirt was picked at random, it would most likely be size 8.

3

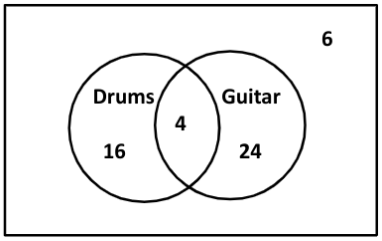
In Year 10 at Harmony High School, there are 50 students. Twenty play drums, 28 play the guitar, and 4 play both.

(a) Draw a Venn diagram to represent this situation.

(b) How many
(i) do not play either instrument?
(ii) only play the drums?
(iii) only play the guitar?
(iv) play at least one instrument?
(v) play at most one instrument?

(c) If one student was picked at random, what is the probability that he/she:
(i) does not play either instrument?
(ii) plays at least one instrument?

(a)



(b) From the Venn diagram:

(i) 6 students don't play either instrument.

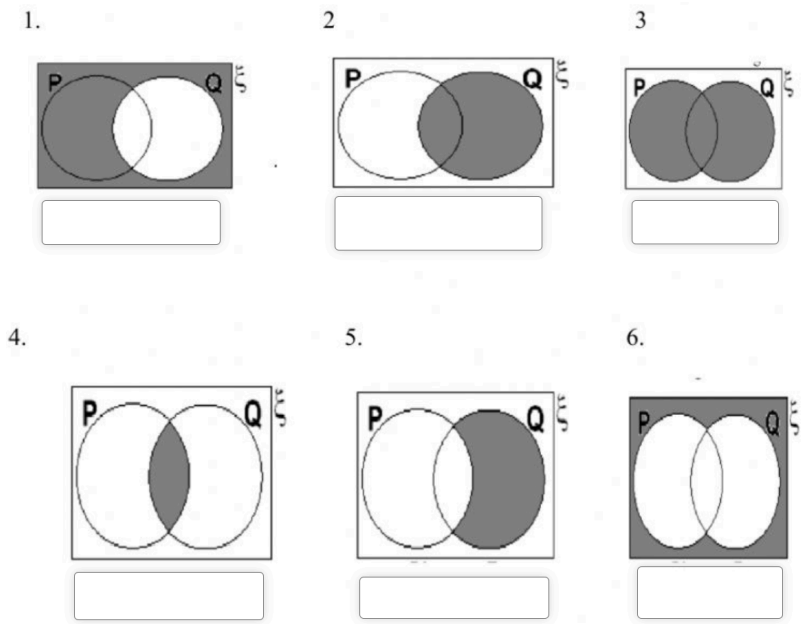
(ii) 16 play only drums

(iii) 24 only play the guitar

(iv) 'At least one' means one or two! So, 16 + 4 + 24 = 44 students play at least one instrument.

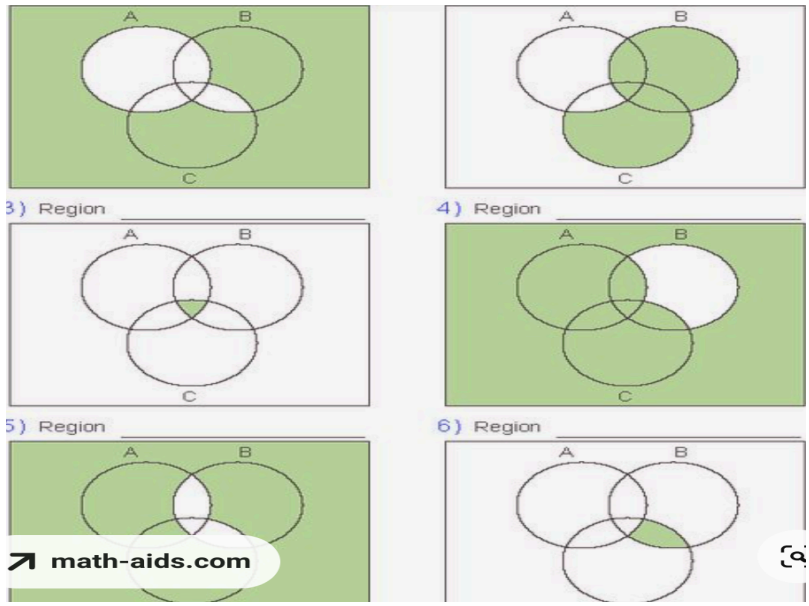
(v) 'At most one' means one or none! So, the only ones who would not be counted are students who play **both** subjects. That is 50 - 4 = 46 students

4. Give the names of the shaded region in the diagrams below:



Several ways to write the shaded regions, here are my solutions

1. Q' 2. Q 3. P U Q 4. P ∩ Q 5.P' ∩ Q 6. (P U Q)'



More than 1 way to do these

1. $(A \cup (B \cap C))'$
2. $(A' \cap C) \cup B$
3. $A \cap B \cap C$
4. $(A \cup C) \cup B'$
5. $(A \cap B)' \cup (B \cap C)'$
6. $(B \cap C) \cap (A \cap B \cap C)'$

5. Nicola did a survey in her Year 10 class, about careers that her colleagues wanted to have when they left school. Of the 30 students in the class, 5 wanted to be teachers, 6 wanted to go into the medical field, 8 into banking, 3 into the transport industry and 8 wanted to be farmers.

(a) Draw up a table to summarise this information.

(b) Calculate relative frequency for each of the choices.

(c) There are 120 students in the whole of Year 10. Using this probability model, how many students could be expected to want to be teachers?

(d) Based on this probability model, if a student in year 10 was chosen at random, what is the probability that he/she would want to:

(i) be a farmer
(ii) **Not** be a farmer
(iii) **not** be a teacher
(iv) go into banking **or** the medical field

a. b.

	Frequency	Relative Frequency
Teacher	5	5/30 = 1/6
Medical	6	6/30 = 1/5
Banking	8	8/30 = 4/15
Transport	3	3/30 = 1/10
Farming	8	8/30 = 4/15

c) Using this probability model, if 1/6 want to be teachers,
1/6 of 120 = 1/6 x 120 = 20

d) (i) P(farmer) = 4/15

(ii) P(**not** farmer) = 1 – 4/15 ('farmer' and '**not** farmer' are complementary events)
= 11/15

(iii) P(**not** teacher) = 1 – 1/6 ('teacher' and '**not** teacher' are complementary events)

(iv) P(banking **or** medical) = 4/15 + 1/5 'banking' and 'medical' are mutually
= 7/15

6. A survey was done with 50 customers in a store to determine the number of people who owned laptop computers and cell phones.

The results were as follows:

- 28 have laptop computers (L)
- 35 have cell phones (C)
- 21 have both L and C

(a) Calculate n(LC **or** MP), the number of customers who have a laptop computer **or** a mobile phone.

(b) Draw a Venn diagram to illustrate the information.

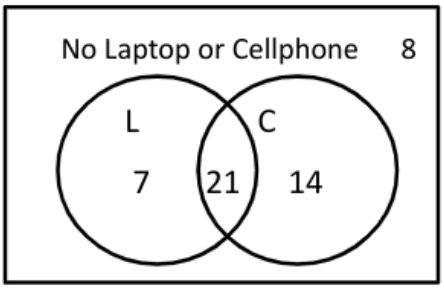
(c) If a customer was chosen at random, what is the probability that he/she has:

(i) a Laptop computer
(ii) a laptop computer, but no cell phone
(iii) neither a laptop computer nor a cell phone

(a) These two events are **not** mutually exclusive, so we must be careful to only count the 'overlap' once!

Ie n(L **or** C) = n(L) + n(C) – n(L **and** C)
= 28 + 35 - 21
= 42

(b)



(c) (i) P(L) = 28/50 = 14/25

(ii) P(L **<u>but no C</u>**) = 7/50

(iii) (no L and no C) = 8/50 = 4/25

Two coins are tossed at the same time.

a Display the sample space in a 2D grid.

b How many different outcomes are possible?

c Find the probability of tossing a heads and a tail.

A coin and a six-sided die are thrown at the same time.

a Display the sample space in a 2D grid.

b How many different outcomes are possible?

7. Possible combinations when two coins are tossed.

a.

Coin 1	H	T	H	T
Coin 2	H	T	T	H

Another way to show it

HH	TT
HT	TH

b. Sample Space S = { HH, TT, HT, TH }, 4 different outcomes

c. Probability of tossing a H and a T : Favorable outcomes are HT and TH
P = 2/4 = ½

8.

Coin↓Die→	1	2	3	4	5	6
H	H1	H2	H3	H4	H5	H6

T	T1	T2	T3	T4	T5	T6

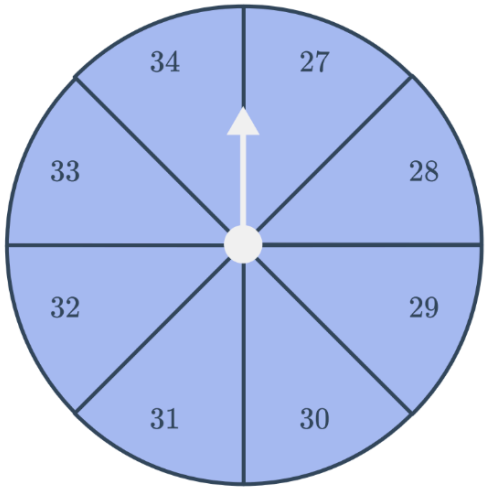
b. Sample Space S = { H1, T1, H2, T2, H3, T3, H4, T4, H5, T5, H6, T6 }, 12 different outcomes.

9. Determine the mean of the numbers 7, 5, 7, 2, 8 and 7. If two additional numbers, 2 and x, reduce the mean by 1, find x.

Mean = 6, x = 2

10.

The following spinner is spun and a six-sided die is rolled at the same time:



a Construct a 2D grid to display the sample space.

b How many different outcomes are possible?

c Find the probability of getting two even numbers.

Spinner

Die↓	27	28	29	30	31	32	33	34
1	1, 27	1, 28	1, 29	1, 30	1, 31	1, 32	1, 33	1, 34
2	2, 27	2, 28	2, 29	2, 30	2, 31	2, 32	2, 33	2, 34
3	3, 27	3, 28	3, 29	3, 30	3, 31	3, 32	3, 33	3, 34
4	4, 27	4, 28	4, 29	4, 30	4, 31	4, 32	4, 33	4, 34
5	5, 27	5, 28	5, 29	5, 30	5, 31	5, 32	5, 33	5, 34
6	6, 27	6, 28	6, 29	6, 30	6, 31	6, 32	6, 33	6, 34

b. Possible Outcomes: 6 x 8 = 48

c. Probability of 2 even numbers: Favorable outcomes: (2, 28), (4,28), (6,28), (2,30), (4,30), (6,30), (2,32), (4,32), (6,32), (2,34), (4,34), (6,34)

P = 12/ 48 = 1/4

11. The number of elements of a set S can be written as n(S).
List the members of the given sets below and state n(S) in each case.
A ={Odd numbers < 12} A = {1,3,5,7,9,11} n(A) = 6
B ={Even numbers ≤ 14} B = {0, 2,4,6,8,10,12,14} n(B) = 8
C ={Square numbers between 20 and 70} C = {25,36,49,64} n(C) = 4
D ={Vowels} D = {a,e,i,o,u} n(D) = 5
E ={Months with 30 days} E = {September, April, June, November} n(E) = 4
F ={prime numbers p; 11≤ p≤ 39} F = {11,13,17,19,23,29,31,37} n(F) = 8
G ={Cube numbers C: 10≤ C≤ 50} G = {27} n(G) = 1

12. Marley rolls two 6-sided dice. Find the probability that he:

	1	2	3	4	5	6
1	1,1	1,2	1,3	1,4	1,5	1,6
2	2,1	2,2	2,3	2,4	2,5	2,6
3	3,1	3,2	3,3	3,4	3,5	3,6
4	4,1	4,2	4,3	4,4	4,5	4,6
5	5,1	5,2	5,3	5,4	5,5	5,6
6	6,1	6,2	6,3	6,4	6,5	6,6

a. Rolls a 2 on both dice.
Rolls a 2 on dice 1 = $\frac{1}{6}$
Rolls a 2 on dice 2 = $\frac{1}{6}$
Rolls a 2 on dice 1 and 2 = $\frac{1}{6}$ x $\frac{1}{6}$ = $\frac{1}{36}$
or as seen in the grid 1 out of 36, 1/36 - (2,2)

b. Rolls a 1 on exactly one of the dice.
Rolls a 1 on dice 1 and anything else on dice 2= $\frac{1}{6}$ x $\frac{5}{6}$ = $\frac{5}{36}$
Rolls a 1 on dice 2 and anything else on dice 1 = $\frac{1}{6}$ x $\frac{5}{6}$ = $\frac{5}{36}$
On Either of the dice = $\frac{5}{36}$ + $\frac{5}{36}$ = $\frac{10}{36}$
or as seen in the grid 10 out of 36, - (1,2), (1,3), (1,4), (1,5), (1,6), (2,1), (3,1), (4,1), (5,1), (6,1)

c. Rolls a 1 or a 2 on either die. (many ways to think about it)

Rolls a 1 on dice 1 and anything else on dice 2 = $\frac{6}{36}$

Rolls a 1 on dice 2 and anything else on dice 1 except 1 = $\frac{5}{36}$

Rolls a 2 on dice 1 and anything else on dice 2 except 1 = $\frac{5}{36}$

Rolls a 2 on dice 2 and anything else on dice 1 except 1 or 2 = $\frac{4}{36}$

$\frac{6}{36} + \frac{5}{36} + \frac{5}{36} + \frac{4}{36} = \frac{20}{36} = \frac{5}{9}$

OR as seen in the grid 20 out of 36,

(1,2), (1,3), (1,4), (1,5), (1,6), (2,1), (3,1), (4,1), (5,1), (6,1), (1,1), (2,2), (2,3), (2,4), (2,5), (2,6), (3,2), (4,2), (5,2), (6,2)

OR

Rolls a 1 on dice 1 and anything else on dice 2 = $\frac{6}{36}$

Rolls a 1 on dice 2 and anything else on dice 1 = $\frac{6}{36}$

Rolls a 2 on dice 1 and anything else on dice 2= $\frac{6}{36}$

Rolls a 2 on dice 2 and anything else on dice 1 = $\frac{6}{36}$

Subtract the double counted (1,2), (2,1), (1,1) & (2,2) = $\frac{4}{36}$

$= \frac{6}{36} + \frac{6}{36} + \frac{6}{36} + \frac{6}{36} - \frac{4}{36} = \frac{20}{36}$

12. There are eight shirts in your closet, four blue and four green. You randomly select one to wear on Monday and then a different one on Tuesday. You wear blue shirts both days.

3/14